CS 229, Fall 2018
Problem Set #0: Linear Algebra and Multivariable Calculus

1. **[0 points] Gradients and Hessians**
Recall that a matrix $A \in \mathbb{R}^{n\times n}$ is *symmetric* if $A^T = A$, that is, $A_{ij} = A_{ji}$ for all i, j. Also recall the gradient $\nabla f(x)$ of a function $f : \mathbb{R}^n \to \mathbb{R}$, which is the n-vector of partial derivatives

$$\nabla f(x) = \begin{bmatrix} \frac{\partial}{\partial x_1} f(x) \\ \vdots \\ \frac{\partial}{\partial x_n} f(x) \end{bmatrix} \qquad \text{where} \qquad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

The Hessian $\nabla^2 f(x)$ of a function $f : \mathbb{R}^n \to \mathbb{R}$ is the $n \times n$ symmetric matrix of twice partial derivatives,

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2}{\partial x_1^2} f(x) & \frac{\partial^2}{\partial x_1 \partial x_2} f(x) & \cdots & \frac{\partial^2}{\partial x_1 \partial x_n} f(x) \\ \frac{\partial^2}{\partial x_2 \partial x_1} f(x) & \frac{\partial^2}{\partial x_2^2} f(x) & \cdots & \frac{\partial^2}{\partial x_2 \partial x_n} f(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2}{\partial x_n \partial x_1} f(x) & \frac{\partial^2}{\partial x_n \partial x_2} f(x) & \cdots & \frac{\partial^2}{\partial x_n^2} f(x) \end{bmatrix}.$$

   (a) Let $f(x) = \frac{1}{2} x^T A x + b^T x$, where A is a symmetric matrix and $b \in \mathbb{R}^n$ is a vector. What is $\nabla f(x)$?
   (b) Let $f(x) = g(h(x))$, where $g : \mathbb{R} \to \mathbb{R}$ is differentiable and $h : \mathbb{R}^n \to \mathbb{R}$ is differentiable. What is $\nabla f(x)$?
   (c) Let $f(x) = \frac{1}{2} x^T A x + b^T x$, where A is symmetric and $b \in \mathbb{R}^n$ is a vector. What is $\nabla^2 f(x)$?
   (d) Let $f(x) = g(a^T x)$, where $g : \mathbb{R} \to \mathbb{R}$ is continuously differentiable and $a \in \mathbb{R}^n$ is a vector. What are $\nabla f(x)$ and $\nabla^2 f(x)$? (*Hint:* your expression for $\nabla^2 f(x)$ may have as few as 11 symbols, including $'$ and parentheses.)

Solution.

(a)
We have

$$f(x) = \frac{1}{2} x^T A x + b^T x,$$

so

$$\nabla f(x) = \frac{1}{2} \nabla (x^T A x) + \nabla (b^T x).$$

First, consider the quadratic term. Let

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad A = (A_{ij})_{n\times n}.$$

Then

$$x^T A x = \sum_{j=1}^{n} \sum_{i=1}^{n} x_i A_{ij} x_j.$$

Fix $k \in \{1, \ldots, n\}$. The terms involving x_k occur either when $i = k$ or when $j = k$. Therefore,

$$x^T A x = \sum_{j=1}^{n} x_k A_{kj} x_j + \sum_{i=1}^{n} x_i A_{ik} x_k - x_k^2 A_{kk} + R(x_k),$$

where $R(x_k)$ denotes all terms that are independent of x_k. The term $x_k^2 A_{kk}$ is subtracted because it appears in both sums. Since A is symmetric, $A_{ki} = A_{ik}$, so

$$x^T A x = 2x_k \sum_{i=1}^{n} A_{ik} x_i - x_k^2 A_{kk} + R(x_k).$$

Separating the $i = k$ term from the summation gives

$$\begin{aligned} x^T A x &= 2x_k \left(\sum_{i \neq k} A_{ik} x_i + A_{kk} x_k \right) - x_k^2 A_{kk} + R(x_k) \\ &= 2x_k \sum_{i \neq k} A_{ik} x_i + x_k^2 A_{kk} + R(x_k). \end{aligned}$$

Hence,

$$\frac{\partial}{\partial x_k}(x^T Ax) = 2\sum_{i\neq k} A_{ik}x_i + 2A_{kk}x_k.$$

Combining the $i = k$ term back into the summation,

$$\frac{\partial}{\partial x_k}(x^T Ax) = 2\sum_{i=1}^{n} A_{ik}x_i.$$

Since

$$(A^T x)_k = \sum_{i=1}^{n} A_{ik}x_i,$$

we obtain

$$\frac{\partial}{\partial x_k}(x^T Ax) = 2(A^T x)_k.$$

Because $A^T = A$,

$$\frac{\partial}{\partial x_k}(x^T Ax) = 2(Ax)_k.$$

Therefore,

$$\nabla(x^T Ax) = 2Ax.$$

Next,

$$b^T x = \sum_{i=1}^{n} b_i x_i,$$

so

$$\frac{\partial}{\partial x_k}(b^T x) = b_k,$$

and therefore

$$\nabla(b^T x) = b.$$

Thus,

$$\nabla f(x) = \frac{1}{2}(2Ax) + b = \boxed{Ax + b}.$$

(b)
Let

$$f(x) = g(h(x)).$$

For each $k \in \{1, \ldots, n\}$, by the chain rule,

$$\frac{\partial}{\partial x_k} g(h(x)) = g'(h(x))\frac{\partial h(x)}{\partial x_k}.$$

Therefore,

$$\frac{\partial f}{\partial x_k} = g'(h(x))\frac{\partial h}{\partial x_k}.$$

Thus,

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = \begin{bmatrix} g'(h(x))\frac{\partial h}{\partial x_1} \\ g'(h(x))\frac{\partial h}{\partial x_2} \\ \vdots \\ g'(h(x))\frac{\partial h}{\partial x_n} \end{bmatrix}.$$

Since $g'(h(x))$ is a scalar,

$$\nabla f(x) = g'(h(x)) \begin{bmatrix} \frac{\partial h}{\partial x_1} \\ \frac{\partial h}{\partial x_2} \\ \vdots \\ \frac{\partial h}{\partial x_n} \end{bmatrix}.$$

Hence,

$$\boxed{\nabla f(x) = g'(h(x))\nabla h(x)}.$$

(c)
We have

$$f(x) = \frac{1}{2}x^T Ax + b^T x, \qquad A^T = A, \qquad b \in \mathbb{R}^n.$$

By definition,

$$\nabla^2 f(x) = \nabla\big(\nabla f(x)\big).$$

From part (a),

$$\nabla f(x) = Ax + b.$$

The ith component of the gradient is

$$(Ax + b)_i = \sum_{k=1}^{n} A_{ik}x_k + b_i.$$

Hence,

$$\frac{\partial f}{\partial x_i} = b_i + \sum_{k=1}^{n} A_{ik}x_k.$$

Now consider the (i, j) entry of the Hessian:

$$\frac{\partial}{\partial x_j}\left(\frac{\partial f}{\partial x_i}\right) = \frac{\partial}{\partial x_j}\left(b_i + \sum_{k=1}^{n} A_{ik}x_k\right) = A_{ij}.$$

Therefore,

$$\boxed{\nabla^2 f(x) = A}.$$

(d)
Let

$$f(x) = g(a^T x),$$

where

$$a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Then

$$a^T x = \sum_{i=1}^{n} a_i x_i.$$

For each i,

$$\frac{\partial f}{\partial x_i} = \frac{\partial g(a^T x)}{\partial x_i} = \frac{\partial g(a^T x)}{\partial (a^T x)} \frac{\partial (a^T x)}{\partial x_i}.$$

Since

$$\frac{\partial (a^T x)}{\partial x_i} = a_i,$$

we obtain

$$\frac{\partial f}{\partial x_i} = g'(a^T x) a_i.$$

Hence,

$$\boxed{\nabla f(x) = g'(a^T x) a}.$$

Now consider the (i, j) entry of the Hessian:

$$\big(\nabla^2 f(x)\big)_{ij} = \frac{\partial}{\partial x_j}\left(\frac{\partial f}{\partial x_i}\right).$$

Using

$$\frac{\partial f}{\partial x_i} = g'(a^T x) a_i,$$

we have

$$\begin{aligned}\left(\nabla^2 f(x)\right)_{ij} &= \frac{\partial}{\partial x_j}\left(g'(a^T x)a_i\right) \\ &= a_i \frac{\partial g'(a^T x)}{\partial (a^T x)} \frac{\partial (a^T x)}{\partial x_j} \\ &= a_i g''(a^T x) a_j \\ &= a_i a_j g''(a^T x).\end{aligned}$$

Since the (i, j) entry of aa^T is $a_i a_j$, we conclude that

$$\boxed{\nabla^2 f(x) = g''(a^T x)\, aa^T}.$$

2. **[0 points] Positive definite matrices**
A matrix $A \in \mathbb{R}^{n\times n}$ is *positive semi-definite* (PSD), denoted $A \succeq 0$, if $A = A^T$ and $x^T Ax \geq 0$ for all $x \in \mathbb{R}^n$.
A matrix A is *positive definite*, denoted $A \succ 0$, if $A = A^T$ and $x^T Ax > 0$ for all $x \neq 0$, that is, all non-zero vectors x.
The simplest example of a positive definite matrix is the identity I (the diagonal matrix with 1s on the diagonal and 0s elsewhere), which satisfies

$$x^T Ix = \|x\|_2^2 = \sum_{i=1}^{n} x_i^2.$$

   (a) Let $z \in \mathbb{R}^n$ be an n-vector. Show that $A = zz^T$ is positive semidefinite.
   (b) Let $z \in \mathbb{R}^n$ be a non-zero n-vector. Let $A = zz^T$. What is the null-space of A? What is the rank of A?
   (c) Let $A \in \mathbb{R}^{n\times n}$ be positive semidefinite and $B \in \mathbb{R}^{m\times n}$ be arbitrary, where $m, n \in \mathbb{N}$. Is BAB^T PSD? If so, prove it. If not, give a counterexample with explicit A, B.

(a) Solution.
Let

$$z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}.$$

Then

$$A = zz^T,$$

so

$$A_{ij} = z_i z_j.$$

Hence,

$$A_{ji} = z_j z_i = z_i z_j = A_{ij},$$

and therefore

$$A^T = A.$$

Now let $x \in \mathbb{R}^n$. Then

$$x^T Ax = x^T zz^T x.$$

Since $z^T x$ is a scalar,

$$x^T zz^T x = (x^T z)(z^T x) = (z^T x)^2.$$

Therefore,

$$x^T Ax = (z^T x)^2 \geq 0.$$

Thus,

$$\boxed{A = zz^T \text{ is positive semidefinite.}}$$

(b) Solution.
Since

$$A = zz^T,$$

for any $x \in \mathbb{R}^n$,

$$Ax = zz^T x = z(z^T x).$$

Therefore,

$$Ax = 0 \iff z^T x = 0,$$

because $z \neq 0$.
Hence,

$$\boxed{\text{Null}(A) = \{x \in \mathbb{R}^n : z^T x = 0\} = z^{\perp}.}$$

Now consider the column space of A. Since

$$Ax = z(z^T x)$$

for every $x \in \mathbb{R}^n$, every vector in $\text{Col}(A)$ is a scalar multiple of z. Thus,

$$\text{Col}(A) \subseteq \text{span}\{z\}.$$

To prove the reverse inclusion, it is enough to show that $z \in \text{Col}(A)$.
Since $z \neq 0$,

$$z^T z = \|z\|_2^2 > 0.$$

Take

$$x = \frac{z}{z^T z}.$$

Then

$$Ax = zz^T \left(\frac{z}{z^T z}\right) = z\frac{z^T z}{z^T z} = z.$$

Therefore,

$$z \in \text{Col}(A).$$

Since $\text{Col}(A)$ is a subspace, it contains every scalar multiple of z. Hence,

$$\text{span}\{z\} \subseteq \text{Col}(A).$$

Therefore,

$$\text{Col}(A) = \text{span}\{z\}.$$

Since $z \neq 0$, the space $\text{span}\{z\}$ has dimension 1. Thus,

$$\boxed{\text{rank}(A) = 1.}$$

(c) Solution.
Yes, BAB^T is positive semidefinite.
Since A is positive semidefinite,

$$A = A^T.$$

Therefore,

$$(BAB^T)^T = BA^T B^T = BAB^T,$$

so BAB^T is symmetric.
Now let $x \in \mathbb{R}^m$. Then

$$x^T(BAB^T)x = (B^T x)^T A(B^T x).$$

Let

$$y = B^T x \in \mathbb{R}^n.$$

Since A is positive semidefinite,

$$y^T Ay \geq 0.$$

Hence,

$$x^T(BAB^T)x \geq 0.$$

Therefore,

$$\boxed{BAB^T \text{ is positive semidefinite.}}$$

3. **[0 points] Eigenvectors, eigenvalues, and the spectral theorem**
The eigenvalues of an $n \times n$ matrix $A \in \mathbb{R}^{n \times n}$ are the roots of the characteristic polynomial

$$p_A(\lambda) = \det(\lambda I - A),$$

which may (in general) be complex.
They are also defined as the values $\lambda \in \mathbb{C}$ for which there exists a vector $x \in \mathbb{C}^n$ such that

$$Ax = \lambda x.$$

We call such a pair (x, λ) an eigenvector, eigenvalue pair.
In this question, we use the notation $\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ to denote the diagonal matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$, that is,

$$\mathrm{diag}(\lambda_1, \ldots, \lambda_n) = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

(a) Suppose that the matrix $A \in \mathbb{R}^{n \times n}$ is diagonalizable, that is,

$$A = T \Lambda T^{-1},$$

for an invertible matrix $T \in \mathbb{R}^{n \times n}$, where

$$\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$$

is diagonal.
Use the notation $t^{(i)}$ for the columns of T, so that

$$T = \begin{bmatrix} t^{(1)} & \cdots & t^{(n)} \end{bmatrix},$$

where $t^{(i)} \in \mathbb{R}^n$.
Show that

$$At^{(i)} = \lambda_i t^{(i)},$$

so that the eigenvalue/eigenvector pairs of A are

$$(t^{(i)}, \lambda_i).$$

A matrix $U \in \mathbb{R}^{n \times n}$ is orthogonal if

$$U^T U = I.$$

The spectral theorem, perhaps one of the most important theorems in linear algebra, states that if $A \in \mathbb{R}^{n \times n}$ is symmetric, that is,

$$A = A^T,$$

then A is diagonalizable by a real orthogonal matrix.
That is, there are a diagonal matrix $\Lambda \in \mathbb{R}^{n \times n}$ and an orthogonal matrix $U \in \mathbb{R}^{n \times n}$ such that

$$U^T A U = \Lambda,$$

or, equivalently,

$$A = U \Lambda U^T.$$

Let

$$\lambda_i = \lambda_i(A)$$

denote the ith eigenvalue of A.

(b) Let A be symmetric.
Show that if

$$U = \begin{bmatrix} u^{(1)} & \cdots & u^{(n)} \end{bmatrix}$$

is orthogonal, where $u^{(i)} \in \mathbb{R}^n$, and

$$A = U \Lambda U^T,$$

then $u^{(i)}$ is an eigenvector of A and

$$Au^{(i)} = \lambda_i u^{(i)},$$

where

$$\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n).$$

(c) Show that if A is PSD, then

$$\lambda_i(A) \geq 0$$

for each i.

(a) Solution.
Since

$$A = T\Lambda T^{-1},$$

multiplying on the right by T gives

$$AT = T\Lambda.$$

Write

$$T = \begin{bmatrix} t^{(1)} & \cdots & t^{(n)} \end{bmatrix}.$$

Then

$$AT = \begin{bmatrix} At^{(1)} & \cdots & At^{(n)} \end{bmatrix}.$$

Since

$$\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n),$$

we also have

$$T\Lambda = \begin{bmatrix} \lambda_1 t^{(1)} & \cdots & \lambda_n t^{(n)} \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} At^{(1)} & \cdots & At^{(n)} \end{bmatrix} = \begin{bmatrix} \lambda_1 t^{(1)} & \cdots & \lambda_n t^{(n)} \end{bmatrix}.$$

Comparing the ith columns gives

$$\boxed{At^{(i)} = \lambda_i t^{(i)}}.$$

Hence $(t^{(i)}, \lambda_i)$ is an eigenvector/eigenvalue pair of A.

(b) Solution.
Since

$$A = U\Lambda U^T$$

and U is orthogonal,

$$U^T U = I.$$

Multiplying $A = U\Lambda U^T$ on the right by U gives

$$AU = U\Lambda.$$

Write

$$U = \begin{bmatrix} u^{(1)} & \cdots & u^{(n)} \end{bmatrix}.$$

Then

$$AU = \begin{bmatrix} Au^{(1)} & \cdots & Au^{(n)} \end{bmatrix},$$

while

$$U\Lambda = \begin{bmatrix} \lambda_1 u^{(1)} & \cdots & \lambda_n u^{(n)} \end{bmatrix}.$$

Comparing the ith columns gives

$$\boxed{Au^{(i)} = \lambda_i u^{(i)}}.$$

Hence $u^{(i)}$ is an eigenvector of A with eigenvalue λ_i.

(c) Solution.
Since A is positive semidefinite,

$$x^T Ax \geq 0$$

for every $x \in \mathbb{R}^n$.

Let $u^{(i)}$ be an eigenvector of A corresponding to the eigenvalue λ_i. Then

$$Au^{(i)} = \lambda_i u^{(i)}.$$

Therefore,

$$(u^{(i)})^T A u^{(i)} = (u^{(i)})^T \lambda_i u^{(i)} = \lambda_i \|u^{(i)}\|_2^2.$$

Since A is PSD,

$$\lambda_i \|u^{(i)}\|_2^2 \geq 0.$$

Because $u^{(i)} \neq 0$,

$$\|u^{(i)}\|_2^2 > 0.$$

Hence,

$$\boxed{\lambda_i(A) \geq 0}$$

for every i.